# Cost-Benefit Evaluation of Seismic Risk Mitigation Alternatives for Older Concrete Frame Buildings

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Cost-benefit assessment is used to evaluate the effectiveness of various seismic retrofit strategies to address concerns about earthquake safety and damage to older concrete frame buildings. The benefits of mitigation are quantified in monetary terms by assessing the reductions in risk of building damage (repairs) and occupant fatalities. When evaluated considering repair costs only, the maximum cost of retrofitting that offers a positive rate of return is 10% to 30% of the building replacement value. Since seismic retrofits generally cost more than this, this result suggests that reduced risk of damage and repairs alone does not warrant the expense of retrofitting. However, the added benefits of reduced fatality risks would make retrofitting cost-effective if it costs as much as 60% of the building replacement cost. The cost-benefit approach provides an important tool to evaluate the merits of seismic mitigation in light of competing demands for finite public and private resources. [DOI: 10.1193/030911EQS040M]

## INTRODUCTION

Nonductile reinforced concrete frame structures are potentially vulnerable to earthquake-induced collapse, threatening the safety of building occupants and incurring significant economic losses. In regions of the United States with high seismic risk, concerns stem from buildings constructed before the mid-1970s, that is, prior to the institution of major changes in seismic design requirements for concrete structures in response to damage from the 1971 San Fernando earthquake. Recent research has shown that these older concrete frame structures are approximately 40 times more susceptible to seismic collapse (in terms of annualized risk) than modern code-conforming concrete frames (Liel et al. 2011). Older concrete buildings are also more likely to be damaged in earthquakes, thus incurring more costly repairs, than their modern counterparts (Liel and Deierlein 2008). Alternatives for mitigating these risks range from structural retrofitting to building demolition and replacement. There are ongoing debates among engineers and policy makers about whether policies should be adopted to encourage or require owners of these structures to undertake risk assessment and mitigation. Emerging performance-based earthquake engineering methods can contribute significant new insights to these discussions.

In this paper, mitigation alternatives for older concrete frame buildings are evaluated through cost-benefit assessment, in which the costs of seismic upgrading or replacement

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are compared to the benefits of these actions. Benefits include reduced risks of fatalities and damage. This study utilizes data from a companion study (Liel et al. 2011, Liel and Deierlein 2008), wherein the authors used seismic performance assessment techniques to calculate and compare the risks of earthquake-induced damage and collapse of 1960s-era concrete frame buildings in California with retrofitted and modern concrete buildings. This paper extends our earlier studies to incorporate cost-benefit analyses to assess the cost-effectiveness of various risk mitigation strategies. The primary objectives of this effort are twofold: first, to illustrate the application of performance-based cost-benefit analyses, and, second, to provide data that inform the debate about retrofitting existing concrete frame buildings. While our focus is on engineering practice and seismic risks for older reinforced concrete buildings in the western United States, this approach can be generally applied to other building types and regions.

## OVERVIEW OF COST-BENEFIT ASSESSMENT

Cost-benefit assessment provides a tool for comparing the costs of a mitigation tactic, such as retrofitting, to the benefits achieved, in this case, from improved seismic performance. By explicitly quantifying the relationship between mitigation effectiveness and its costs, these assessments facilitate effective decision making for investment in seismic safety. Such efforts respond to state and local government directives, like those in California's Multi-Hazard Mitigation Plan, which encourage "cost-effective long term mitigation" activities to reduce seismic risk (OES 2010, p. 32).

Cost-benefit assessment is used here to evaluate the costs and benefits of retrofitting or replacing potentially vulnerable older concrete frame buildings. The costs represent the expenditure needed to retrofit or rebuild a building. Benefits come from the improved seismic performance, that is, the reduced risks of fatalities or damage of the retrofitted or replaced building. For comparison with the retrofit or replacement costs, the benefits are quantified in monetary terms. The economic evaluation of life-safety benefits requires assigning a dollar value to human life, which is controversial but necessary to evaluate the economic effectiveness of seismic risk mitigation measures relative to the mitigation of other risks in the face of finite resources for public safety. The implied value of human life, or the amount society is willing to pay to save a life, depends on the nature of the risk, the victim's remaining lifespan, future earning potential, and other factors. In engineering and construction projects, human life is typically valued in the range of $2 to $5 million (Mitrani-Reiser 2007), which is about the same as the average amount awarded per victim by the September 11$^{th}$ Victim Compensation Fund (Feinberg et al. 2004). Interestingly, these values are substantially lower than those associated with health regulations for some contaminants such as arsenic, which have been estimated to cost as much as $100 million per life saved (Haimes 2008).

The benefits of seismic mitigation are realized sometime in the future, if and when an earthquake occurs. Uncertainty in the timing, location, and severity of future earthquakes is accounted for through the seismic hazard information for a particular building site and the assumed Poisson distribution model of earthquake occurrence. Future benefits are quantified in terms of equivalent annualized values and discounted to present-day values for comparison with the up-front costs of mitigation. Present-value discounting accounts for the time-value of money, recognizing that money paid or earned today is valued more than the same amount in

the future. The discount rate is determined from interest rates and adjusted for inflation, and traditionally ranges from 2% to 6% (Nuti and Vanzi 2003). Following Pate-Cornell (1984), the same discount rate is applied to the dollar benefits associated with lives and economic losses.

This study builds on prior research that applied cost-benefit methods to evaluate seismic design and mitigation alternatives. Previous work includes cost-benefit assessments evaluating the cost-effectiveness of: (1) new designs that exceed code-minimum seismic design requirements (Wen and Kang 2001a and 2001b, Takahashi et al. 2004), (2) seismic upgrades for infrastructure systems and bridges (Kunreuther et al. 2001, Padgett et al. 2010, Nuti and Vanzi 2003), and (3) seismic retrofitting of unreinforced masonry and concrete industrial buildings in Boston (Pate-Cornell 1985), wooden houses in California (Porter et al. 2006), and existing concrete buildings in Turkey and Greece (Smyth et al. 2004, Kappos and Dimitrakopoulos 2008). These studies show the cost-effectiveness of retrofitting to be highly dependent on the cost of the retrofit, the level of strengthening, the seismicity of the region, and the time horizon considered. The studies also differ in the level of detail applied in the risk assessment. In this regard, this study is distinguished from prior ones by employing detailed nonlinear structural analyses and the latest performance-based earthquake engineering methods to calculate probabilistic performance metrics, including building-specific estimates of economic losses and fatalities due to earthquake damage and collapse. These methods are applied to examine the cost-effectiveness of seismic retrofitting to older concrete frame buildings, focusing on typical configurations and retrofitting techniques that are representative of construction practice in the United States.

## PERFORMANCE-BASED COST-BENEFIT ASSESSMENT METHODOLOGY

Performance-based cost-benefit assessment begins with analysis of the seismic performance of archetypical existing concrete buildings, along with the retrofitted and replaced buildings that represent the various mitigation alternatives. These analyses employ nonlinear structural models capable of capturing strength and stiffness degradation in earthquake-damaged buildings. Each building model is subjected to a set of ground-motion time histories through the process of incremental dynamic analysis, wherein each ground motion is scaled in increasing intensity until the onset of structural collapse (Vamvatsikos and Cornell 2002).

Incremental dynamic analysis is used to calculate probabilistic metrics of structural response, damage and collapse risk. The structural collapse capacity is defined by a lognormal probability distribution that describes the probability of collapse as a function of the ground-motion intensity, $P[Collapse|im]$ (Zareian and Krawinkler 2007, Haselton et al. 2011). Statistically, this lognormal distribution is described by a median collapse intensity and a logarithmic standard deviation representing the dispersion in the ground-motion intensity causing collapse. The dispersion accounts for variability in the ground-motion records and modeling uncertainties; the typical value of about 0.6 also approximately represents the coefficient of variation in the collapse capacity (Liel et al. 2009). The mean annual frequency of collapse (i.e., the collapse rate per year) is obtained by the integration of the structure's collapse capacity distribution with the seismic hazard curve,

$$\lambda_{collapse} = \int_{IM=0}^{\infty} P[Collapse|im]|d\lambda_{IM}| \tag{1}$$

where $d\lambda_{IM}$ is the derivative of the ground-motion hazard curve, which describes the likelihood of ground motions occurring that exceed a particular intensity at the building site. Details about the structural analyses and collapse performance assessments for the nonductile concrete frame buildings are described in Liel et al. (2008 and 2011).

Earthquake-induced economic losses are calculated by estimating damage to a building's nonstructural and structural components and then determining the aggregate cost of repairing this damage (Miranda et al. 2004, Porter and Kiremidjian 2001, ATC 2011, Ramirez et al. 2012). Damage measures are defined for an individual component based on the repair actions needed to restore it to its undamaged state. Component fragility functions then relate levels of deformation or acceleration the component experiences during an earthquake to the probability a component will experience a certain damage state. Structural losses are aggregated from those experienced by all components, and computed for every level of ground-motion intensity of interest. Following Ramirez et al. (2012), the total expected annualized loss (EAL) is obtained by combining the expected losses, $E[L|im]$, associated with the collapse and non-collapse ($NC$) state of the building, integrated over all ground-motion intensities:

$$EAL = \int_{IM=0}^{\infty} E[L|im]|d\lambda_{IM}| \tag{2a}$$

$$E[L|im] = E[L|NC, im](1 - P[Collapse|im]) + E[L|Collapse, im]P[Collapse|im] \tag{2b}$$

EAL represents the estimated losses, in terms of an average yearly amount, associated with repairing earthquake damage, considering the frequency and severity of possible future earthquakes represented by the seismic hazard at the site of interest.

The next step of the performance assessment is to calculate the risk of loss of human life by combining information on whether or not a collapse occurs with assumptions about the portion of the building that collapses, the number of people likely to have been in the collapsed part of the building, and factors that affect how lethal collapse is for building occupants (Mitrani-Reiser 2007, Coburn et al. 1992). The expected annual rate of fatalities (EAF), which represents an average yearly number of fatalities, accounting for all possible future earthquakes, is computed similarly to the annualized losses:

$$EAF = \int_{IM=0}^{\infty} E[F|Collapse, im]P[Collapse|im]|d\lambda_{IM}| \tag{3}$$

The seismic performance assessments for the original and retrofitted or replacement buildings are then used to determine the benefits of each mitigation alternative. The expected benefit of the retrofit or replacement action, $E[B]$, is given by:

$$E[B] = E[B_F] + E[B_L] \tag{4}$$

where $B_F$ are benefits due to lives saved (reduced fatalities) and $B_L$ are benefits due to avoided repairs costs (reduced economic losses). Specifically, benefits associated with reduced economic losses are computed from the difference in expected annual losses for the retrofitted ($EAL_R$) and the original ($EAL_o$) buildings. The expected present value of benefits over the building's remaining lifespan is given by the following:

$$E[B_L] = (EAL_o - EAL_R) \sum_{t=1}^{T} (1 + r)^{-t} \tag{5a}$$

$$E[B_F] = (EAF_o - EAF_R) V \sum_{t=1}^{T} (1 + r)^{-t} \tag{5b}$$

In these computations, a constant discount rate, $r$, of 3% and a remaining building life, $T$, of 50 years are assumed. The dollar benefits associated with improved safety depend on the value of each life saved, $V$, which is taken here as \$2 million.

The outcome of the cost-benefit assessment is the ratio of mitigation costs to the expected benefits. The costs of retrofitting and replacement are determined from standard cost-estimating databases and information from practicing engineers. Cost-benefit ratios less than unity indicate favorable conditions, where the benefits outweigh the costs.

## SEISMIC PERFORMANCE OF ARCHETYPE CONCRETE FRAME BUILDINGS

This study evaluates mitigation alternatives for older concrete frame buildings of the type built before significant seismic design and detailing provisions became required for concrete moment frame design in the mid-1970s. Seismic performance of these older buildings is determined through assessment of a set of archetypical concrete office buildings designed according to the 1967 *Uniform Building Code*, or 1967 UBC (ICBO 1967). The buildings range in height from 2 to 12 stories (28 to 158 ft.), include both space and perimeter frames, and are designed for Seismic Zone 3 according to the 1967 UBC, which at the time was the highest seismic zone, covering most of California. Buildings have an 8-inch floor slab and plan areas of 125 × 175 ft. (2- and 4-story buildings) or 125 × 125 ft. (8- and 12-story), with columns spaced at 25 ft. on center. Typical story heights are 13 ft., except for first stories, which are 15 ft. tall. Design and detailing of the structures is intended to be representative of California's older concrete moment frame buildings, excluding buildings with infill walls or major strength or stiffness irregularities. The buildings are assumed to be located at a representative stiff-soil site in Los Angeles with high seismic risk, for which probabilistic seismic hazard analysis is available (Goulet et al. 2007). Liel et al. (2011 and 2008) reports further details of the structural designs and the performance assessments summarized below.

Key metrics of seismic performance for the archetype older buildings, which provide the basic input data for the cost-benefit assessment, are summarized in Table 1. Nonlinear building response was simulated with OpenSees software (OpenSees 2012), using lumped plasticity models of beams, columns, and beam-column joints. Column and beam plastic

**Table 1.** Metrics of earthquake-induced collapse risk, economic losses and fatalities in archetype concrete frame structures

| | Older concrete frame buildings | | | | | Modern concrete frame buildings | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Bldg ID[a] | $T_1$[b] (sec) | Collapse[c] capacity (g) | $\lambda_{collapse}$[d] ($\times 10^{-4}$) | EAL[e] (%) | EAF[f] ($\times 10^{-3}$) | $T_1$ (sec) | Collapse capacity (g) | $\lambda_{collapse}$ ($\times 10^{-4}$) | EAL (%) | EAF ($\times 10^{-3}$) |
| 2S | 1.1 | 0.47 | 109 | 5.2 | 41 | 0.63 | 3.6 | 1.0 | 1.0 | 0.4 |
| 2P | 1.0 | 0.68 | 47 | 3.2 | 24 | 0.66 | 2.5 | 3.4 | 1.0 | 1.7 |
| 4S | 2.0 | 0.27 | 107 | 2.3 | 62 | 0.94 | 2.2 | 1.7 | 1.1 | 1.3 |
| 4P | 2.0 | 0.31 | 100 | 2.3 | 97 | 1.1 | 1.6 | 3.6 | 1.2 | 2.7 |
| 8S | 2.2 | 0.29 | 64 | 1.8 | 75 | 1.8 | 1.2 | 2.4 | 1.3 | 3.1 |
| 8P | 2.4 | 0.23 | 135 | 2.1 | 141 | 1.7 | 1.0 | 6.3 | 1.0 | 8.3 |
| 12S | 2.3 | 0.29 | 50 | 1.6 | 76 | 2.1 | 0.83 | 4.7 | 1.1 | 9.4 |
| 12P | 2.8 | 0.24 | 119 | 1.6 | 192 | 2.1 | 0.85 | 5.2 | 0.77 | 9.9 |

[a] Denotes the number of stories (2, 4, 8, or 12) and the framing system (space, S, or perimeter, P).
[b] First-mode period from eigenvalue analysis of the structural model, accounting for cracked concrete sections.
[c] Median collapse capacity of structural model, in terms of spectral acceleration at the first-mode period, $Sa(T_1)$.
[d] Mean annual frequency of collapse, (i.e., collapses per year).
[e] Expected annual losses, expressed as a percentage of the total replacement cost of the structure.
[f] Expected annual fatalities.

hinge models are calibrated to experimental data to represent columns that are flexurally-dominated, but may subsequently fail in shear (Liel et al. 2011). In Table 1, the structural collapse resistance is represented by the median collapse capacity (specified in terms of ground-motion intensity parameter, $Sa(T_1)$, that is, spectral acceleration at the first-mode period) and the mean annual frequency of collapse ($\lambda_{collapse}$). Modeling earthquake recurrence by a Poisson distribution following the usual assumptions, the computed frequencies of collapse (ranging between 47 to 135 $\times$ $10^{-4}$ collapses per year) equate to a 20% to 50% probability of collapse in 50 years.

Economic losses are also reported for each building in Table 1. The details of these computations, including fragility functions and repair costs associated with different components are described elsewhere (Ramirez et al. 2012, Liel and Deierlein 2008, Mitrani-Reiser 2007). Economic losses associated with repairs are annualized over the lifetime of the structure to give expected annual loss (computed through Equation 2), which vary from 1.6% to 5.2% of the total replacement cost of the building. Predicted annualized seismic fatality rates range from about 0.02 to 0.2 deaths per year.

Seismic performance metrics for modern concrete frames are also reported in Table 1. These results provide a benchmark against which to contrast the performance of the older buildings and a basis for the study of building replacement. The modern office structures were designed according to 2005 code provisions for the Los Angeles site (ACI 2005, ASCE 2005, ICC 2003). Details of the nonlinear simulation models, seismic collapse, economic loss, and fatality assessments for these structures are reported in Haselton et al. (2011), Ramirez et al. (2012), and Liel and Deierlein (2008), respectively.

Compared to metrics obtained for older concrete frames, the modern structures show significantly better seismic performance in terms of reduced collapse and life-safety risks and moderately reduced economic risks. The probabilities of collapse in 50 years are reduced by a factor of 10 to 80 times, ranging from 20% to 50% for the older buildings to about 0.5% to 3% for modern buildings. The decrease in fatality risks is comparable. On the other hand, the economic losses (repair costs) are only about two times smaller for the modern buildings (EALs ranging from 0.8% to 1.3% of replacement value) as compared to the nonductile buildings.

## REPLACEMENT OF OLDER CONCRETE FRAME BUILDINGS

The most drastic mitigation strategy is to replace vulnerable older concrete frame buildings with new structures that fully comply with modern building code requirements. To evaluate this alternative, the costs of demolishing and replacing the 1967-era buildings can be compared with the benefits according to the methodology described above.

### Replacement Costs

The costs of replacing the older structures with modern office buildings are obtained using the RS Means *Costs Estimating Handbook*, which provides a systematic approach for estimating building construction costs (Balboni 2007). Estimated replacement costs (in 2006 dollars) range from $\$140/ft^2$ to $\$170/ft^2$, depending on the height of the building. The estimates account for the cost of constructing all significant structural and nonstructural components in Los Angeles, including HVAC systems, partitions, interior finishes, exterior enclosures, services, and basements for the taller buildings. In addition, the estimated replacement costs include a 25% allowance for contractor overhead and profit. However, the replacement costs do not include additional project costs related to demolition, permitting, administration, management or financing, which may increase construction costs by an additional 50% to 100%. Therefore, the estimates provide a lower bound on the actual cost of replacement.

### Cost-Benefit Assessment

Data comparing the benefits and costs of building replacement are summarized in Table 2. The benefits are described by: (1) the present value of direct economic losses avoided (based on reduction of earthquake-related repair costs), (2) the number of lives saved, and (3) the present value sum of the benefits for reduced fatality and repair cost losses over the assumed 50-year remaining building lifespan. Replacing an older concrete frame building is predicted to (a) save between \$2.7 and \$6.5 million in reduced seismic repair costs and (b) protect two to nine lives, combining to (c) an equivalent present value total benefits of \$4.8 to \$15.6 million. In dollar terms, 40% to 60% of the total benefits come from the safety improvements.

Of the buildings considered, only the replacement of the older 2-story space frame structure has a favorable cost-benefit ratio less than one, where the benefits exceed costs. The older 2-story building has unusually large predicted economic losses, such that, if replaced with a new building, the \$6.5 million present value in savings from reducing earthquake-induced damage and repairs alone covers the cost of replacement. For the other structures,

**Table 2.** Benefits and costs of replacing older concrete frame structures

| | Benefits over 50 years | | | | |
|---|---|---|---|---|---|
| Bldg ID | Economic losses avoided ($, millions) | Lives saved | Total benefits ($, millions) | Cost of building replacement ($, millions) | Cost-benefit ratio |
| 2S | 6.5 | 2.0 | 8.6 | 6.1 | 0.7 |
| 2P | 3.7 | 1.1 | 4.8 | 6.5 | 1.4 |
| 4S | 3.9 | 3.1 | 7.0 | 12.5 | 1.8 |
| 4P | 3.5 | 4.7 | 8.3 | 12.0 | 1.4 |
| 8S | 2.7 | 3.7 | 6.5 | 19.9 | 3.1 |
| 8P | 5.3 | 6.6 | 12.1 | 19.4 | 1.6 |
| 12S | 4.2 | 3.3 | 7.6 | 29.1 | 3.8 |
| 12P | 6.3 | 9.1 | 15.6 | 28.1 | 1.8 |

the computed cost-benefit ratios exceed the break-even value of one, indicating that the replacement costs are not justified by the benefits. The cost-benefit assessment may also be expressed in terms of how much we would need to be willing to spend to obtain the benefits associated with replacement. From this perspective, the total present value benefits of $8.6 million associated with replacement of the 2-story space frame building are worth approximately $\$200/ft^2$, which is more than the estimated replacement cost. On the other hand, the $7.0 million benefits of replacing the 4-story space frame structure are worth only $\$80/ft^2$, which is substantially less than the true cost of replacing the building and, therefore, not cost-effective.

The cost-benefit ratios depend on a number of parameters, including the dollar value assigned to lives saved, the time horizon considered, and the assumed discount (interest) rate. Figure 1 shows how the cost-benefit ratio varies as a function of the cost per life

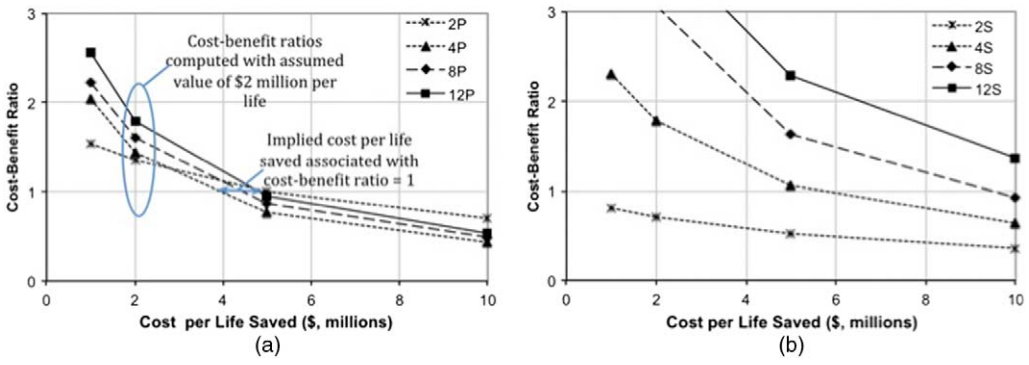



**Figure 1.** Cost-benefit assessment of replacing older concrete structures of different heights for (a) perimeter and (b) space frames, illustrating the impact of the value placed on human life.

saved, where the ratios for life valued at $2 million are the same as those reported in Table 2. Figure 1 can be interpreted as a calculation of the implied cost of saving each human life by replacing the vulnerable older concrete buildings with new buildings, considering the replacement costs and the reductions in direct economic losses, reported in Table 2. To achieve the break-even cost-benefit ratio of one, the implied costs per life saved are between $4 and $6 million per life for the perimeter frame buildings (Figure 1a), and vary from less than $1 million to more than $10 million per life for the space frame buildings (Figure 1b). The larger variability of values in the space frame buildings reflects the larger inherent variability in their performance; the greater costs per life saved in the taller buildings represents the smaller relative reduction in economic losses possible with replacement of those buildings.

Figure 2 shows how investment time horizon and interest rate affect the outcome of cost-benefit assessment for a typical building. If the remaining service life is longer than the assumed 50 years, the benefits of replacement increase, thereby reducing (improving) the cost-benefit ratio. However, in most cases solely extending the service life will not make the investments in seismic improvement pay off, because the present value of future benefits saturates. Even more critical is the observation that shorter time spans escalate the cost-benefit ratio, making the investment much less attractive. For existing older buildings, these shorter time horizons may, in fact, be more realistic for the performance-based cost-benefit assessment since these buildings may be close to the end of their useful life and there is frequent ownership turnover in the U.S. real estate market. Figure 2 also illustrates that when the real interest rates increase, the present value of future benefits is reduced and the cost-benefit ratio increases. The higher interest rates effectively make this mitigation alternative less attractive than other investments.

Taken together, the data in Table 2 and Figures 1 and 2 demonstrate the large variability in cost-benefit assessment results due to the inherent variability in building designs and

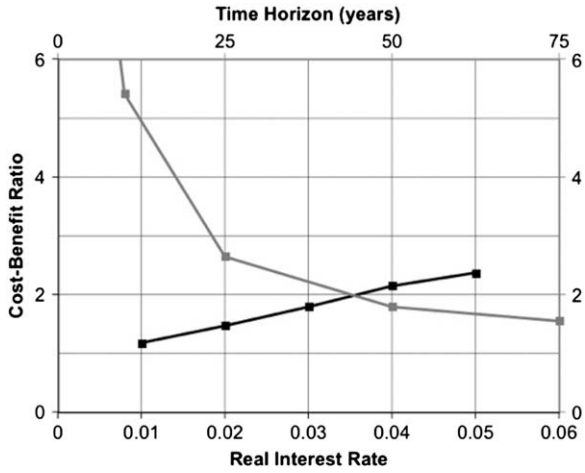



**Figure 2.** Effect of time horizon (gray) and real interest rate (black) on cost-benefit assessment of replacing the 12-story older concrete perimeter frame building.

performance combined with uncertainties in the input parameters. Therefore, it is difficult to draw general conclusions as to the cost-effectiveness of seismic risk mitigation for the general class of older concrete buildings. It should be emphasized that since all the existing buildings considered here have regular configurations and are code-conforming (albeit to outdated requirements), they likely represent older concrete buildings with average or above performance. A large number of existing older concrete frame buildings would likely have irregularities due to architectural constraints or infill walls, errors and omissions in design or construction, material deterioration, or soft-soil sites that, if accounted for, increase predicted seismic-related losses and fatalities. Cost-benefit ratios for building replacement would be more favorable in those cases.

## RETROFIT OF OLDER CONCRETE FRAME BUILDINGS

Compared to replacement, seismic retrofit is likely to be a more realistic and cost-effective choice for seismic mitigation. Strategies to retrofit older concrete frames include the addition of concrete shear walls or steel braces or the jacketing of columns with steel, concrete, or fiber polymer products. Depending on how they are designed, these methods can reduce deformation demands by stiffening or strengthening the structure, and/or inhibit brittle failure modes by improving ductility at the member or system level (ASCE 2007, Bai 2003). Retrofit design for an existing building depends on the site's seismic hazard, the building use and occupancy, the desired seismic performance level, as well as any specific structural deficiencies. For reasons of cost and practicality, retrofits are typically not expected to match the performance of new buildings and, as a result, the improvements in seismic performance after retrofitting may vary significantly from building to building (ASCE 2007, FEMA 2009). Retrofit designs are generally intended to at least meet minimum life-safety or collapse-prevention goals for rare (maximum considered) earthquake ground motions.

### Retrofit Designs and Models

As illustrated in Figure 3, three alternative seismic retrofitting techniques are evaluated here: (1) jacketing existing columns with reinforced concrete, (2) adding wall piers, and (3) wrapping existing columns with fiber-reinforced composites. These techniques are applied to the archetype 4- and 8-story older concrete frame buildings, which were prone to collapse due to a combination of premature story mechanisms and limited column and beam plastic hinge ductility. Two levels of retrofit were considered for each technique. The "minimum" retrofit level represents the minimum practical intervention that would be applied for each seismic retrofit technique, while the "significant" retrofits reflect more extensive enhancements. In each case, the retrofitted elements are designed and detailed in accordance with standard practice. Retrofit design characteristics are then used to modify nonlinear dynamic analysis models to assess the behavior of the retrofitted building. Seismic performance of the retrofitted structures is then quantified in terms of risk of economic losses, collapse, and fatalities, following the same procedures as applied to the original building designs.

Note that in this study, retrofits are designed without regard for the peculiar seismic deficiencies of an individual archetype building. As such, the study examines a generic set of retrofitting schemes, rather than any specific building, retrofit method, or detailed design. The retrofits are intended to represent ordinary, rather than extraordinary, efforts to improve seismic performance while minimizing the impact on building architecture and layout.

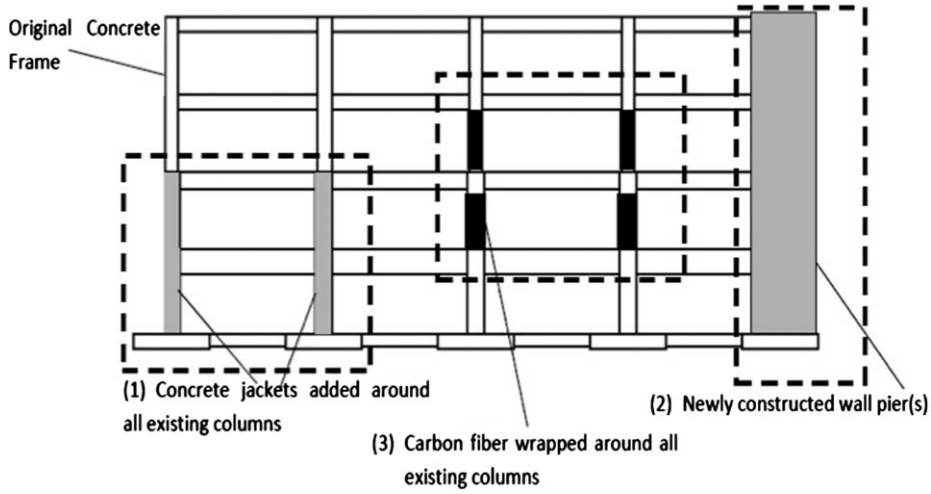



**Figure 3.** Illustration of retrofit strategies considered for archetype older concrete frame buildings.

*Concrete Jacket Retrofits*

In this strategy, the buildings are retrofitted by encasing all existing columns in concrete. These jackets increase the strength, stiffness, and ductility of nonductile columns, thereby enhancing their deformation capacity, increasing the ratio of column-to-beam strength and improving the strength and confinement of beam-column joints. Table 3 describes the jacketed column designs, denoted RCJ-M and RCJ-S for the minimum and significant retrofits. Jacketed concrete column specimens from past research (Bousias et al. 2007a and 2007b, Rodriguez and Park 1994, Julio et al. 2005) were used to determine values for jacket thickness and amounts of transverse and longitudinal reinforcement. Minimum retrofits correspond to the smallest feasible jacket size, consisting of new concrete measuring 3 inches thick around the column. Significant retrofits are designed with thicker jackets and enough longitudinal reinforcement to increase column flexural strength to achieve a ratio of column-to-beam strength that exceeds 1.2 at every joint. This strength ratio is equivalent to the ACI 318 strong column/weak beam requirement for modern concrete frames. Transverse reinforcement in jackets is spaced at approximately 5 inches, providing nearly twice the confining reinforcement than is present in unretrofitted columns, and similar levels of confinement to code-conforming concrete columns.

Composite action of the original column and new jacket is incorporated in simulation models to represent strength, stiffness, and deformation capacity of jacketed members (Bousias et al. 2007b). The concrete jacket and added longitudinal reinforcement strengthen the columns by 60% to 130% and stiffen the frames by 30% to 140%. The new transverse rebar roughly doubles the deformation capacity of the original nonductile columns. The concrete jackets also enhance the strength and confinement of beam-column joints, and the joint shear models are modified accordingly. The nonlinear static pushover analysis results in Figure 4 show the significant increase in the stiffness, strength, and ductility of the older

**Table 3.** Key design and model properties for retrofitted concrete frame structures

| Bldg ID | Retrofit/mitigation description | $T_1$ (sec) | Column/wall dimensions[a,b] | Column plastic rotation capacity, $\theta_{cap,pl}$ $(rad)$[e] |
|---|---|---|---|---|
| 4S | Original (No retrofit) | 2.0 | 20 × 20 in. | 0.021 |
| | RCJ-M | 1.5 | 26 × 26 in. | 0.054 |
| | RCJ-S | 1.3 | 28 × 28 in. | 0.056 |
| | W-M | 1.5 | 36 × 16 in. | 0.035 |
| | W-S | 1.1 | 72 × 16 in. | 0.039 |
| | CFRP | 2.0 | 20 × 20 in. | 0.053 |
| | Replacement | 0.9 | 22 × 22 in.[c] | 0.057 |
| 8S | Original (No retrofit) | 2.2 | 28 × 28 in. | 0.013 |
| | RCJ-M | 1.9 | 34 × 34 in. | 0.045 |
| | RCJ-S | 1.9 | 34 × 34 in.[d] | 0.045 |
| | W-M | 1.7 | 72 × 16 in. | 0.032 |
| | W-S | 1.5 | 144 × 16 in. | 0.036 |
| | CFRP | 2.2 | 28 × 28 in. | 0.033 |
| | Replacement | 1.8 | 22 × 22 in. | 0.051 |

[a] Properties reported for interior first-story columns.
[b] For wall retrofits, the wall size is reported. Wall sizes are constant over the height of the building. Dimensions here are the true wall size. Model wall properties are scaled for inclusion in the 2-D frame models.
[c] Column size and strength are not comparable between the older buildings (unretrofitted or retrofitted) and the modern (replacement) buildings. The modern structures have 20-ft. (not 25-ft.) column spacing and more columns.
[d] For this structure, differences between RCJ-M and RCJ-S are not apparent in interior first-story column.
[e] Plastic rotation capacities are a key modeling parameter representing component ductility. Parameter values reported are based on the modeled monotonic backbone curve and shown for interior first-story columns or the base of the wall. Values for the CFRP retrofit include the confining effect of the fiber wrap.

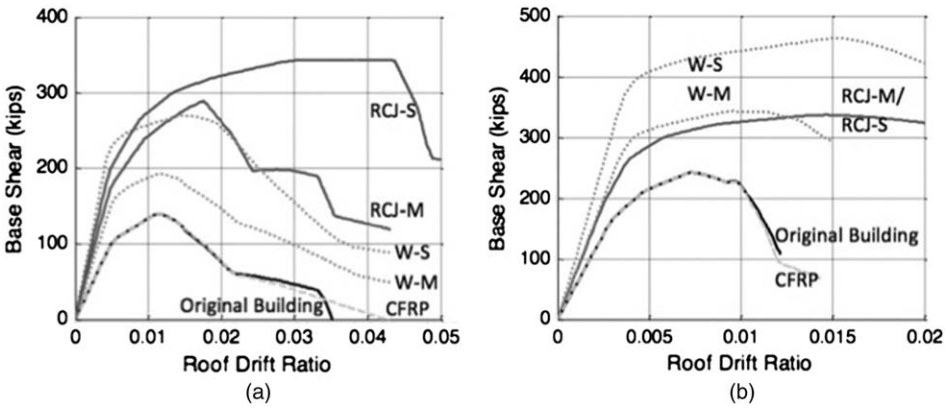



**Figure 4.** Results of pushover analysis for retrofitted (a) 4-story and (b) 8-story concrete frame buildings.

concrete building due to jacketing. Much of this improvement results from the inhibition of story mechanisms by the strengthened columns and beam-column joints.

*Shear Wall Retrofits*

In this retrofit, concrete shear walls are added to strengthen and stiffen the original frame. These walls could be constructed as infill by enlarging existing columns, as shown in Figure 3. Two walls are added to each exterior building face. The minimal shear wall retrofits (W-M) are designed to increase the lateral strength of the original frame (as obtained from pushover analysis) by 40% and the significant retrofits (W-S) are designed to double the lateral strength.

Wall design and model features are summarized in Table 3. Wall length varies from 3 to 12 ft., and the walls are designed to prevent shear failures, such that their behavior is flexurally dominated. Retrofit shear walls meet the detailing and boundary zone requirements in the ACI 318-05 seismic provisions, with stirrups spaced at 4 inches. Model parameters (such as plastic rotation capacity) are based on the predictive equations also used to model concrete columns (Haselton et al. 2008). Following Fardis and Biskinis (2003), the predicted deformation capacity is reduced by 40%, to account for the relatively less ductile behavior of walls as compared to columns. Wall elements are modeled as large fixed-base columns, recognizing that this level of foundation fixity may be difficult to achieve in practice. As shown in the pushover results (Figure 4), the wall retrofit is somewhat less effective than column jacketing for the 4-story building and more effective for the 8-story building.

*Carbon Fiber Wrap Retrofits*

In the final retrofit application, the nonductile concrete columns of the frame are wrapped with a carbon fiber reinforced polymer composite (CFRP), as shown in Figure 3. CFRP is used primarily to improve column shear strength and, to a lesser extent, column confinement and ductility, reflecting standard practice in the United States (Bakis et al. 2002, Seible et al. 1997, Liel 2011). The beams and joints are not wrapped because this is rarely done, due to complications and costs associated with providing continuity and anchorage through the joint regions. Column flexural strength is not affected, as this would require additional CFRP plies with fibers running parallel to the column's longitudinal axis and anchorage of the longitudinal plies into the joint region.

The CFRP retrofits are designed to prevent shear failure in columns. In addition, the CFRP is assumed to improve the deformation capacity and energy dissipation model parameters of the column plastic hinge regions in simulation models, as reported in Table 3. This assumption is supported by experimental data for CFRP-wrapped rectangular concrete columns, which showed a 60% to 250% increase in deformation capacity (Galal et al. 2005, Iacobucci et al. 2003, Sause et al. 2004). The analysis models do not include fracture or debonding of the fiber-wrap since these failure modes are assumed to be prevented through design and detailing. Referring to Figure 4, the CFRP has only a modest effect on the pushover response of the buildings because it does not mitigate the formation of premature story mechanisms associated with weak columns.

### Seismic Performance of Retrofitted Concrete Frame Buildings

Table 4 summarizes the seismic performance assessment of the retrofitted concrete frames; Figure 5 shows collapse fragility curves for each building. As expected, retrofitting decreases the predicted collapse rates, though the extent of the improvement depends on the retrofit. The concrete jacket retrofits reduce the 50-year collapse probability of 4- and 8-story buildings from 28% and 41% (for unretrofitted buildings) down to between 3% and 7% (for retrofitted buildings), respectively. For these retrofits, greater column and joint strength reduce flexural hinging in columns, distributing damage over the height of the building. Closely spaced transverse reinforcement in the jackets improves column confinement and deformation capacity. The wall retrofits reduce the 50-year collapse probability to between 4% and 13%. The addition of a ductile shear wall to the lateral resisting system stiffens the structure and alters the collapse mechanisms in comparison to the unretrofitted frames, leading to multistory mechanisms and significant energy dissipation in the upper stories. While the shear walls reduce deformation demands on existing columns, they do not improve their ductility, which is evident in the steeper post-peak pushover response of the wall retrofits than the jacketed retrofits, as shown in Figure 4. The fiber wrap retrofits were only marginally

**Table 4.** Effect of retrofitting older concrete frame buildings, including seismic performance assessment (metrics of collapse risk, economic losses, and fatalities) and cost-benefit assessment results

| | | | | | Benefits over 50 years | | | |
|---|---|---|---|---|---|---|---|---|
| Retrofit description | Collapse capacity (g) | $\lambda_{collapse}$ ($\times 10^{-4}$) | EAL (%) | EAF ($\times 10^{-3}$) | Economic losses avoided ($, millions) | Lives saved | Total benefits ($, millions) | Break-even retrofit cost ($/ft$^2$) |
| *4-story concrete space frames* | | | | | | | | |
| Original | 0.27 | 107 | 2.3 | 62 | n/a | n/a | n/a | n/a |
| RCJ-M | 0.89 | 14 | 1.2 | 17 | 3.2 | 2.3 | 5.6 | 64 |
| RCJ-S | 1.2 | 6.9 | 1.2 | 8.8 | 3.5 | 2.7 | 6.2 | 71 |
| W-M | 0.62 | 27 | 1.8 | 35 | 1.7 | 1.3 | 3.1 | 35 |
| W-S | 0.93 | 16 | 1.9 | 22 | 1.2 | 2.0 | 3.3 | 38 |
| CFRP | 0.35 | 75 | 2.1 | 50 | 0.6 | 0.6 | 1.1 | 13 |
| Replacement | 2.2 | 1.7 | 1.1 | 1.3 | 3.9 | 3.1 | 7.0 | n/a |
| *8-story concrete space frames* | | | | | | | | |
| Original | 0.29 | 64 | 1.8 | 77 | n/a | n/a | n/a | n/a |
| RCJ-M | 0.84 | 7.3 | 1.5 | 10 | 1.5 | 3.3 | 4.9 | 40 |
| RCJ-S | 0.92 | 6.4 | 1.3 | 8.5 | 2.5 | 3.4 | 6.0 | 48 |
| W-M | 0.66 | 20 | 1.5 | 34 | 1.4 | 2.2 | 3.6 | 29 |
| W-S | 0.97 | 9.3 | 1.4 | 17 | 1.9 | 3.1 | 5.0 | 40 |
| CFRP | 0.38 | 60 | 1.6 | 68 | 1.1 | 0.5 | 1.5 | 12 |
| Replacement | 1.2 | 2.4 | 1.3 | 3.1 | 2.7 | 3.7 | 6.5 | n/a |

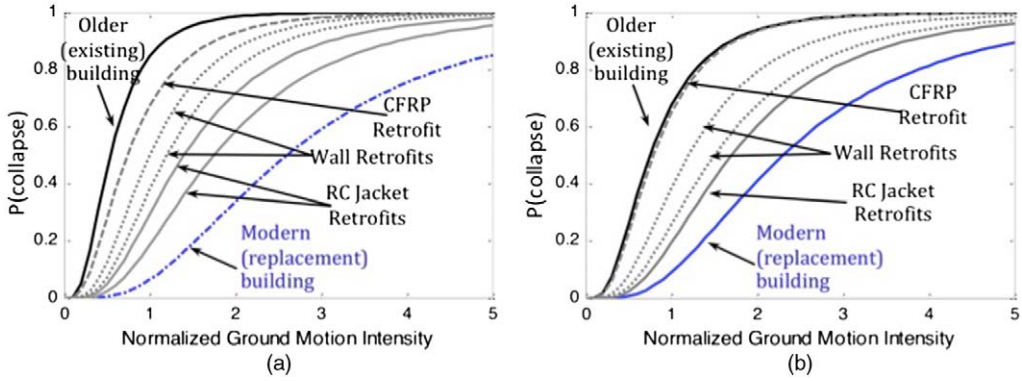



**Figure 5.** Collapse fragility functions for (a) 4-story space frames and (b) 8-story space frames. Note: The ground-motion intensity is normalized by the spectral acceleration that has 2% chance of exceedance in 50 years at the site of interest to permit comparison among structures with different first-mode periods.

effective at reducing collapse risk for the 4- and 8-story frames. While the applied CFRP prevents column shear failure, the columns of these frames are not particularly shear-critical in most cases. Likewise, the retrofits increase column confinement, but do not reduce the tendency for soft-story collapse because the relative strength of columns, beams, and joints in the frames is unaltered. CFRP is likely more effective at reducing risk for buildings with deficient lap-splice connections, or those with columns, walls, or coupling beams requiring additional shear strength.

The retrofitted structures are generally safer, and in some cases considerably safer, than their unretrofitted counterparts. Annualized fatality rates (reported in Table 4) in the retrofitted buildings are a factor of 1.1 to 9.0 times lower than those in the original buildings. Of the retrofit strategies examined, the column jacket and shear wall retrofits are most effective at improving safety, although none of the retrofit strategies reduce earthquake-related fatalities to the level of safety in the new (replacement) buildings designed in accordance with modern building codes.

Retrofitting the older concrete frames decreases the expected annual losses by a factor of 1.1 to 1.9. Of the retrofit techniques evaluated, the concrete jacket retrofits have the most significant impact on losses, because the increase in frame stiffness lessens damage and losses in drift-sensitive structural and nonstructural components. For the same reason, wall retrofits also reduce repair costs and losses. Even so, retrofit leads to a larger reduction in life safety risks (on a relative basis) than in direct repair cost losses. As such, the reduction in human safety risks presents the primary motivation for seismic rehabilitation.

None of the retrofit strategies performed as well as modern concrete frame structures. In practice, some building-specific retrofits probably do achieve code-conforming levels of structural performance, by targeting higher performance levels or carefully addressing specific deficiencies. For example, more significant improvements in seismic performance of jacketed retrofitted structures are possible with larger jackets, and the performance of

wall retrofits would improve with greater wall strength/stiffness. The evaluation also illustrates the variability in seismic performance of retrofitted structures due to the differences in aggressiveness of the retrofit, performance goals, and the quality of design and construction.

### Retrofit Costs

The cost of retrofitting older concrete buildings varies substantially, depending on the extent of the rehabilitation, characteristics of the existing structure, the materials used, labor and time required for construction, the disruption of usable space, the level of architectural renovation accompanying seismic work, and the historic value of the building. Due to the large variability in retrofitting costs we did not attempt to estimate retrofitting costs on a case-by-case basis, instead we used published data on retrofitting costs and information from practicing engineers to define the expected range of costs of retrofitting older concrete frame structures. FEMA 156 reports typical costs of seismic rehabilitation, depending on the structural system, size of structure, site conditions and seismic hazard, and retrofit performance goal (FEMA 1994). For a concrete frame building in Los Angeles on a site with "very high" seismicity, the estimated cost of seismic upgrade ranges between $\$35/ft^2$ and $\$70/ft^2$, depending on the level of the retrofit (2006 dollars). These appraisals are relatively consistent with estimates of costs of $\$20/ft^2$ to $\$100/ft^2$ for retrofitting concrete frames, obtained from practicing engineers in California (Liel 2011).

### Costs and Benefits of Retrofitting

The results of the cost-benefit assessment for the retrofitted buildings are summarized in the final four columns of Table 4. As in the cost-benefit assessment of building replacement, the benefits of retrofitting, in dollar-terms, are due to the reduction in earthquake-related repair costs and fatalities. Since retrofitting costs are unknown, the outcome of the assessment is the value of the retrofit performance benefits (in $\$/ft^2$), reported in the last column of Table 4 and compared in Figure 6. This amount is the maximum break-even cost, implying that if the retrofit can be constructed for this price, it is cost-effective. The CFRP retrofits cannot be justified for the archetype buildings from a cost-benefit standpoint since they are

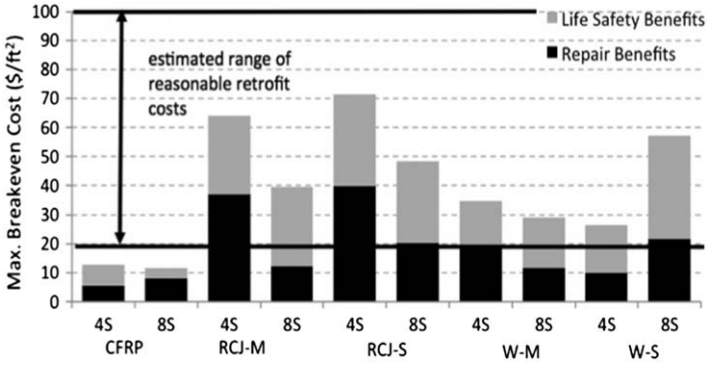



**Figure 6.** Maximum cost of different retrofit strategies to achieve neutral cost-benefit assessment.

valued at only $\$11/ft^2$ to $\$13/ft^2$, less than even the minimum of the range of retrofit costs. Several of the other retrofit strategies, including concrete jacket and some wall retrofits, are valued at $\$40/ft^2$ to $\$70/ft^2$, which are in the range of being cost-effective, relative to expected retrofit costs.

On average, approximately 50% of the computed benefits come from saving lives and the balance from reduced economic losses. If the benefits to public safety are neglected, the savings in direct losses is generally not sufficient to warrant most of these retrofits from a cost-benefit perspective. For example, imagine that the RCJ-S retrofit for the 4-story older concrete frame can be achieved for $\$80/ft^2$, for a total cost of \$6.9 million (55% of the building replacement cost). This retrofit reduces future economic losses, corresponding to a \$3.5 million benefit. The remaining cost of \$3.4 million (present value) could be justified by lives saved over the next 50 years, equivalent to about \$2.4 million per life saved. If the same retrofit could be achieved more cheaply, suppose for only $\$45/ft^2$, the cost per life saved would reduce to only \$0.2 million because reduced damage and repair cost losses make up for most of the retrofit cost.

It should be noted that this study probably understates the benefits of retrofitting. In practice, thoughtful design of retrofit solutions would further improve their effectiveness and reduce costs to improve the attractiveness of retrofit alternatives. The benefits also do not account for reduction in the downtime or losses associated with building closure while inspections and repairs are carried out (Comerio 2006), the losses associated with damage to building contents or nonfatal injuries.

## CONCLUSIONS

This study investigates the effectiveness of seismic mitigation alternatives for older (nonductile) concrete frame buildings through performance-based cost-benefit assessment. Costs of retrofitting and replacement are estimated using standard cost databases. The benefits of retrofitting or replacing older concrete frames are quantified in terms of the future reductions in earthquake-related repairs and improvements in safety (reduced fatalities). These benefits are computed from the results of robust nonlinear simulations and assessments of earthquake-induced losses and fatalities on a building-by-building basis. Using standard techniques from engineering economics, future benefits are converted to expected present value for comparison with costs, which are incurred today.

Replacement of existing concrete buildings with new code-conforming buildings is shown to significantly improve public safety (reducing collapse rates by a factor of about 40) and reduce seismic repair costs (by a factor of about two). Despite these significant improvements, the cost-benefit assessment of building replacement indicates that the costs exceed the benefits in most cases for the assumed values of interest rate, value of human life, and remaining building lifespan. This outcome is attributed to the large costs associated with constructing a new building and the relatively small reduction in repair costs achieved. As a result, the implied cost per life saved is approximately \$5 million.

The observation that safety dominates the cost-evaluation for replacing older concrete frame buildings supports the idea of a limited retrofit that primarily aims to prevent collapse and loss of life. The cost-effectiveness of retrofitting older concrete buildings is highly dependent on the cost of retrofitting, which is difficult to estimate. However, this analysis

shows that if the retrofits considered here can be achieved for less than about 50% of the price of building replacement (within the range shown in Figure 6), the cost-benefit ratio outcomes are favorable. Viewed in terms of the implied cost per life saved, retrofitting costs about $2 million per life saved, rather than the $5 million estimated for replacement. It is expected that building-specific retrofits designed by practicing engineers would be even more successful at reducing losses and lives lost, implying that the true benefits of retrofitting may be more significant than those estimated here.

This study provides a systematic and quantitative method for comparing costs and benefits of retrofitting and provides representative results that are applicable to California-type existing building construction in high seismic areas. These metrics of risk, costs, and benefits provide important information for policy-makers dealing with vulnerable concrete buildings in the building stock and building designers and owners deciding between retrofitting alternatives. In the future, results can be extended to consider a wider variation of building types and regions of different seismicity, though the trends identified here are expected to be generally applicable. The tools required to estimate losses and fatalities as needed in the performance-based cost-benefit assessment are increasingly available to researchers and practicing engineers through the ATC-58 project and others (ATC 2011).

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## REFERENCES

American Concrete Institute (ACI), 2005. *Building Code Requirements for Structural Concrete, ACI 318*, Farmington Hills, MI.

American Society of Civil Engineers (ASCE), 2005. *Minimum Design Loads for Buildings and Other Structures, ASCE 7-05*, Reston, VA.

American Society of Civil Engineers (ASCE), 2007. *Seismic Rehabilitation of Existing Buildings, ASCE/SEI 41-06*, Reston, VA.

Applied Technology Council (ATC), 2011. *Guidelines for Seismic Performance Assessment of Buildings ATC-58 75% Draft*, prepared for Federal Emergency Management Agency, Redwood City, CA.

Bai, J.-W., 2003. *Seismic Retrofit of Reinforced Concrete Building Structures*, Mid-America Earthquake Center, Texas A&M University, College Station, TX.

Bakis, C. D., Bank, L. C., Brown, V. L., Cosenza, E., Davalos, J., Lesko, J., Machida, A., Rizkalla, S., and Triantafillou, T., 2002. Fiber-reinforced polymer composites for construction: State-of-the-art review, *Journal of Composites for Construction* **6**, 73–87.

Balboni, B. (ed.), 2007. *Means Square Foot Costs*, R.S. Means Co., Inc., Kingston, MA, 472 pp.

Bousias, S., Spathis, A.-L., and Fardis, M. N., 2007a. Seismic retrofitting of columns with lap spliced smooth bars through FRP or concrete jackets, *Journal of Earthquake Engineering* **11**, 653–674.

Bousias, S. N., Biskinis, D. E., Fardis, M. N., and Spathis, A.-L., 2007b. Strength, stiffness and cyclic deformation capacity of concrete jacketed members, *ACI Structural Journal* **104**, 521–531.

Coburn, A., Spence, R., and Pomonos, A., 1992. Factors determining human casualty levels in earthquakes: Mortality prediction in building collapse, *10th World Conference on Earthquake Engineering*, 19–24 July 1992, Madrid, Spain.

Comerio, M. C., 2006. Estimating downtime in loss modeling, *Earthquake Spectra* **22**, 349–365.

Fardis, M. N., and Biskinis, D. E., 2003. Deformation capacity of RC members, as controlled by flexure or shear, in *Otani Symposium*, University of Tokyo, Japan, 511–530.

Feinberg, K. R. E., Biros, C. S., Feldman, J. H. E., Greenspan, D. E. E., and Zins, J. E. E., 2004. *Final Report of the Special Master for the September 11th Victim Compensation Fund of 2001*, U.S. Department of Justice, Washington, D.C.

Federal Emergency Management Agency (FEMA), 1994. *Typical Costs for Seismic Rehabilitation of Existing Buildings, FEMA 156, 2nd ed.*, Washington, D.C.

Federal Emergency Management Agency (FEMA), 2009. *Engineering Guidelines for Incremental Seismic Rehabilitation, FEMA P-420*, prepared by Applied Technology Council, Redwood City, CA.

Galal, K., Arafa, A., and Ghobarah, A., 2005. Retrofit of RC square short columns, *Engineering Structures* **27**, 801–813.

Goulet, C. A., Haselton, C. B., Mitrani-Reiser, J., Beck, J. L., Deierlein, G. G., Porter, K. A., and Stewart, J. P., 2007. Evaluation of the seismic performance of a code-conforming reinforced-concrete frame building: From seismic hazard to collapse safety and economic losses, *Earthquake Engineering and Structural Dynamics* **36**, 1973–1997.

Governor's Office of Emergency Services (OES), 2010. *State of California Multi-Hazard Mitigation Plan*, Mather, CA.

Haimes, Y. Y., 2008. *Risk Modeling, Assessment and Management, 3rd ed.*, John Wiley & Sons, Hoboken, NJ.

Haselton, C. B., Liel, A. B., and Deierlein, G. G., 2011. Seismic collapse safety of reinforced concrete buildings: I. Assessment of ductile moment frames, *Journal of Structural Engineering* **137**, 481–491.

Haselton, C. B., Liel, A. B., Taylor Lange, S., and Deierlein, G. G., 2008. *Beam-Column Element Model Calibrated for Predicting Flexural Response Leading to Global Collapse of RC Frame Buildings, PEER Report 2007/03*, Pacific Earthquake Engineering Research (PEER) Center, University of California, Berkeley.

Iacobucci, R. D., Sheikh, S. A., and Bayrak, O., 2003. Retrofit of square concrete columns with carbon fiber-reinforced polymer for seismic resistance, *ACI Structural Journal* **100**, 785–794.

International Council of Building Officials (ICBO), 1967. *Uniform Building Code*, Pasadena, CA.

International Code Council (ICC), 2003. *International Building Code*, Falls Church, VA.

Julio, E., Branco, F., and Silva, V. D., 2005. Reinforced concrete jacketing: Interface influence on monotonic loading response, *ACI Structural Journal* **102**, 252–257.

Kappos, A. J., and Dimitrakopoulos, E. G., 2008. Feasibility of pre-earthquake strengthening of buildings based on cost-benefit and life-cycle cost analysis, with the aid of fragility curves, *Natural Hazards* **45**, 33–54.

Kunreuther, H., Cyr, C., Grossi, P., and Tao, W., 2001. Using cost-benefit analysis to evaluate mitigation for lifeline systems, *MCEER Research Progress and Accomplishments 2000-1*, MCEER, Buffalo, NY.

Liel, A. B., and Deierlein, G. G., 2008. *Assessing the Collapse Risk of California's Existing Reinforced Concrete Frame Structures: Metrics for Seismic Safety Decisions, Technical Report No. 166*, Blume Earthquake Engineering Center, Stanford University, Stanford, CA.

Liel, A. B., Haselton, C. B., Deierlein, G. G., and Baker, J. W., 2009. Incorporating modeling uncertainties in the assessment of seismic collapse risk of buildings, *Structural Safety* **31**, 197–211.

Liel, A. B., Haselton, C. B., and Deierlein, G. G., 2011. Seismic collapse safety of reinforced concrete buildings: II. Comparative assessment of non-ductile and ductile moment frames, *Journal of Structural Engineering* **137**, 492–502.

Liel, A. B., 2011. Communication with professional engineers J. Heintz (of the Applied Technology Council, Redwood City, CA), J. Maffei and W. Holmes (of Rutherford and Chekene Engineers, San Francisco, CA), and E. Reis (of Certus Consulting, Oakland, CA).

Miranda, E., Aslani, H., and Taghavi, S., 2004. Assessment of seismic performance in terms of economic losses, in *Proceedings, International Workshop on Performance-Based Seismic Design: Concepts and Implementation*, Pacific Earthquake Engineering Research (PEER) Center, University of California, Berkeley, 149–160.

Mitrani-Reiser, J., 2007. *An Ounce of Prevention: Probabilistic Loss Estimation for Performance Based Earthquake Engineering*, Ph.D. Dissertation, California Institute of Technology, Pasadena, CA.

Nuti, C., and Vanzi, I., 2003. To retrofit or not to retrofit, *Engineering Structures* **25**, 701–711.

Open System for Earthquake Engineering Simulation (OpenSees), 2012. Open-source software framework for simulating the seismic response of structural and geotechnical systems, opensees.berkeley.edu.

Padgett, J. E., Dennemann, K., and Ghosh, J., 2010. Risk-based seismic life-cycle cost-benefit (LCC-B) analysis for bridge retrofit assessment, *Structural Safety* **32**, 165–173.

Pate-Cornell, M. E., 1985. Costs and benefits of seismic upgrading of some buildings in the Boston area, *Earthquake Spectra* **1**, 721–740.

Pate-Cornell, M. E., 1984. Discounting in risk analysis: Capital vs. human safety, in *Proceedings of Symposium on Structural Technology and Risk*, Institute for Risk Research, University of Waterloo, Ontario, Canada, 17–32.

Porter, K. A., and Kiremidjian, A. S., 2001. *Assembly-Based Vulnerability of Buildings and its Uses in Seismic Performance Evaluation and Risk Management Decision-Making, Technical Report No. 309*, Blume Earthquake Engineering Center, Stanford University, Stanford, CA.

Porter, K. A., Scawthorn, C. R., and Beck, J. L., 2006. Cost-effectiveness of stronger woodframe buildings, *Earthquake Spectra* **22**, 239–266.

Ramirez, C. M., Liel, A. B., Mitrani-Reiser, J., Haselton, C. B., Spear, A. D., Steiner, J., Deierlein, G. G., and Miranda, E., 2012. Expected earthquake damage and repair costs in reinforced concrete frame buildings, *Earthquake Engineering and Structural Dynamics* **41**, 1455–1475.

Rodriguez, M., and Park, R., 1994. Seismic load tests on reinforced concrete columns strengthened by jacketing, *ACI Structural Journal* **91**, 150–159.

Sause, R. H., Harries, K., Walkup, S. L., Pessiki, S., and Ricles, J. M., 2004. Flexural behavior of concrete columns retrofitted with carbon fiber-reinforced polymer jackets, *ACI Structural Journal* **101**, 708–716.

Seible, F., Priestley, M. J. N., Hegemier, G. A., and Innamorato, D., 1997. Seismic retrofit of RC columns with continuous carbon fiber jackets, *Journal of Composites for Construction* **1**, 52–62.

Smyth, A., Altay, G., and Deodatis, G., 2004. Probabilistic benefit-cost analysis for earthquake damage mitigation: Evaluating measures for apartment houses in turkey, *Earthquake Spectra* **20**, 171–203.

Takahashi, Y., Der Kiureghian, A., and Ang, A., 2004. Life-cycle cost analysis based on a renewal model of earthquake occurrences, *Earthquake Engineering and Structural Dynamics* **33**, 859–880.

Vamvatsikos, D., and Cornell, C. A., 2002. Incremental Dynamic Analysis, *Earthquake Engineering and Structural Dynamics* **31**, 491–514.

Wen, Y. K., and Kang, Y. J., 2001a. Minimum building life-cycle cost design criteria I. Methodology, *Journal of Structural Engineering* **127**, 330–337.

Wen, Y. K., and Kang, Y. J., 2001b. Minimum building life-cycle cost design criteria II. Applications, *Journal of Structural Engineering* **127**, 338–346.

Zareian, F., and Krawinkler, H., 2007. Assessment of probability of collapse and design for collapse safety, *Earthquake Engineering and Structural Dynamics* **36**, 1901–1914.

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